Group Anagrams

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The problem

- Group words that are anagrams of each other (key = sorted letters).

Intuition

- Anagrams share the same sorted letters, which makes a perfect hash key.

Approach

- 1. Sort each word's letters
- 2. Group words by that key
- 3. Return the groups

Java solution

```java
public class GroupAnagrams {
    public static List<List<String>> groupAnagrams(List<String> words) {
        Map<String, List<String>> groups = new LinkedHashMap<>();
        for (String w : words) {
            char[] c = w.toCharArray();
            Arrays.sort(c);
            groups.computeIfAbsent(new String(c), k -> new ArrayList<>()).add(w);
        }
        return new ArrayList<>(groups.values());
    }
}
```

Complexity

- Time: O(w * n log n)
- Space: O(w * n)

Recap

- Anagrams share the same sorted letters, which makes a perfect hash key.
- Complexity: O(w * n log n)